

# Surgi-Diff: A Diffusion-Based Generative Framework for Postoperative Outcome Prediction and Surgical Planning in Adolescent Idiopathic Scoliosis

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# Surgi-Diff: A Diffusion-Based Generative Framework for Postoperative Outcome Prediction and Surgical Planning in Adolescent Idiopathic Scoliosis

## Abstract

Preoperative surgical planning for patients with adolescent idiopathic scoliosis (AIS) is inherently complex and remains highly dependent on clinical experience. Existing approaches primarily focus on predicting scalar surgical parameters, but lack imaging-level evidence that can support surgical decision-making. To address this gap, we propose Surgi-Diff, a diffusion-based generative framework for postoperative outcome prediction and surgical planning in AIS. By integrating structural priors from preoperative spinal X-rays and surgical region masks through a Spine Alignment Encoder (SAE), Surgi-Diff generates anatomically consistent postoperative X-rays and simultaneously predicts implant-related structures. The framework was developed and evaluated on a retrospective cohort of 612 AIS patients. Experimental results showed that Surgi-Diff achieved the highest SSIM and competitive PSNR among compared methods, with qualitative results demonstrating strong agreement with ground truth (GT) postoperative morphology and implant layout.These results indicate that Surgi-Diff has potential clinical value for AIS preoperative planning, postoperative visualization, and surgical decision support.

**Keywords:** Adolescent idiopathic scoliosis, surgical planning, diffusion model, postoperative outcome prediction, deep learning.

## 1. Introduction

Adolescent idiopathic scoliosis (AIS) is a common spinal deformity that can cause progressive trunk imbalance, cosmetic deformity, pain, and, in severe cases, cardiopulmonary impairment [1]. For progressive AIS, posterior corrective fusion surgery remains the standard treatment for restoring spinal alignment and improving quality of life [2]. However, preoperative planning for AIS is highly complex [3] and strongly dependent on surgical experience, as surgeons must determine fusion levels, correction strategy, and implant configuration while balancing deformity correction and global spinal alignment [4-5]. This process is inherently complex, time-consuming, and highly dependent on surgical expertise, often leading to substantial inter-surgeon variability. Therefore, there is a strong clinical need for automatic tools that can support objective, accurate, and clinically actionable surgical planning.

In current practice, AIS surgical planning is still largely based on radiographic measurements and manually derived planning parameters [6]. Such assessments are laborious and often show substantial inter-rater variability. With recent advances in artificial intelligence (AI), particularly deep learning (DL), automated methods have increasingly been explored for medical image analysis and clinical decision support. In the spine domain, prior studies have mainly addressed deformity assessment [7], progression prediction [8], and postoperative outcome evaluation [9]. A smaller body of work has investigated preoperative planning tasks more directly, including fusion-level selection [10], screw sizing [11], and trajectory planning [12]. However, most existing methods focus on predicting individual parameters in isolation. While these outputs may be clinically informative, they do not adequately represent the

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overall expected surgical outcome and therefore provide limited support for comprehensive planning. In addition, parameter-only predictions lack intuitive visual evidence, which reduces the interpretability and practical value of model outputs in real clinical decision-making.

From a surgical planning perspective, postoperative morphology prediction at the image level is particularly valuable because it can provide a direct visual representation of the expected correction result and instrumentation outcome before surgery [13]. Such a capability would allow surgeons not only to anticipate postoperative spinal alignment, but also to derive planning-relevant information from the predicted image, including fusion extent, screw placement and screw distribution. However, this is a highly challenging task. AIS correction surgery induces substantial non-rigid anatomical deformation, and the postoperative X-ray reflects both large-scale changes in spinal curvature and local structural alterations caused by metallic implants. Consequently, direct same-modality translation from preoperative to postoperative X-rays is far from trivial. When mainstream generative models are applied without explicit structural constraints, they often fail to preserve global spinal alignment and fine anatomical boundaries simultaneously, resulting in structural blurring, implausible artifacts, and reduced anatomical fidelity [14]. These limitations substantially undermine their utility for surgical planning.

To address these challenges, we propose Surgi-Diff, a diffusion-based generative framework for postoperative outcome prediction and surgical planning in AIS. Rather than predicting only isolated numerical indicators, Surgi-Diff formulates surgical planning support as an imaging-level postoperative prediction problem. The framework incorporates a Spine Alignment Encoder (SAE) to extract structural priors from preoperative posteroanterior (PA) spinal X-rays and surgical region masks, thereby guiding the generation of anatomically consistent postoperative spinal X-rays under conditions of large non-rigid correction. In parallel, the framework predicts implant-related structures, including screw and rod regions, enabling downstream estimation of clinically meaningful planning parameters. In this way, Surgi-Diff establishes a bridge between postoperative image synthesis and practical surgical planning by transforming predictive imaging outputs into actionable decision support.

### 1.1 Research challenges

This study addresses two core challenges for AIS surgical planning and postoperative imaging prediction. First, postoperative image generation is jointly determined by preoperative spinal morphology, the planned surgical region, and the selected correction strategy. This requires the model to explicitly leverage surgical-region in order to generate targeted and clinically relevant predictions. Second, practical AIS planning demands not only anatomically faithful postoperative images that reflect realistic surgical outcomes, but also and structured spatial information of implant-related anatomy, including screws and rods, from which quantitative planning parameters can be derived to support actionable surgical decision-making. Therefore, the central challenge is to develop an integrated framework that jointly performs postoperative image generation, implant prediction, and clinical parameter estimation while keeping the spinal anatomy realistic and structurally correct.

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### 1.2 Research contributions

The core contributions of this work are as follows:

1. To the best of our knowledge, this is the first study to explore and develop an image-level postoperative outcome prediction framework for AIS surgical planning. The generated results provide intuitive present of anticipated surgical outcomes, improved model interpretability, and support extraction of important surgical parameters to guide the surgical planning process.
2. We propose a **SAE** to encode structural priors from preoperative PA spinal X-rays and surgical region masks. This design captures global alignment and morphological information, enabling anatomically consistent postoperative X-ray generation despite large non-rigid spinal deformation.
3. The proposed framework produces *clinically actionable outputs* and is comprehensively validated. From the predicted images, clinically relevant planning parameters, including fusion level, rod morphology, and screw distribution, can be automatically derived. Quantitative experiments and expert evaluation demonstrate the potential utility and reliability of the method for AIS surgical planning.

### 1.3 Manuscript organization

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed method, including conditioning, postoperative X-ray synthesis, instrumentation prediction, and surgical parameter computation. Section 4 reports experimental results and analysis. Finally, Section 5 concludes the paper.

## 2. Related Work

### 2.1 Spinal surgery planning and implant prediction

The core objective of spinal deformity correction surgery is to achieve effective reconstruction of the spinal force line through precise implant planning (such as pedicle screws, corrective rods). Traditional planning relies on the physician's experience, which limits reproducibility and introduces substantial subjectivity. Earlier computational approaches, such as statistical shape models and finite element analysis, provided biomechanical insight but were often computationally demanding and highly dependent on handcrafted clinical priors, limiting their routine clinical applicability.

More recently, DL has enabled data-driven approaches for spine surgery planning. Scherer et al. [15] developed and validated an automatic planning tool for lumbar-sacral pedicle screws using Convolutional Neural Networks (CNN), automatically generating screw trajectories (including entry points and directions) from Computed Tomography (CT) images. The verification showed non-inferiority to manual planning and significantly reduced planning time. Subsequently, the U-Net architecture was widely used for implant segmentation. Yang et al. [16] proposed a multi-scale U-Net model, achieving a positioning accuracy of 98.8% for pedicle screws in postoperative X-rays, with high clinical accuracy, providing a quantitative basis for surgical effect evaluation. Liu et al. [17] further verified the accuracy of navigation-assisted technology in pedicle screw placement in their research. By combining robots and navigation systems, real-time monitoring and adjustment of the screw placement process were achieved, improving surgical safety.

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Despite these advances, most existing methods address isolated tasks, such as screw trajectory planning, implant localization, or navigation assistance, rather than the overall problem of preoperative surgical planning. They typically operate on a single modality and do not provide an intuitive prediction of postoperative spinal morphology together with implant layout. As a result, their ability to support holistic decision-making, patient-specific planning, and surgeon-patient communication remains limited.

### 2.2 Generative models for scoliosis imaging analysis

Generative models have created new opportunities for medical image synthesis, augmentation, and predictive imaging, and have increasingly been explored in scoliosis-related applications. Early work in this area was dominated by Generative Adversarial Networks (GANs). For example, He et al. [18] employed a conditional GAN (cGAN) to generate scoliosis X-ray images and further synthesized virtual X-ray images from smartphone photographs to assist Cobb angle assessment. However, GAN-based methods often suffer from training instability, blurred details, and limited anatomical consistency in generated images.

Diffusion models have recently emerged as a more powerful alternative because of their stable optimization and superior image quality. Hung *et al.* [19] applied a latent diffusion model to medical image synthesis and showed improved realism and finer structural preservation in spinal X-ray generation compared with GAN-based approaches. More broadly, diffusion-based methods have demonstrated strong performance in medical image generation, especially in settings requiring high-fidelity structural reconstruction.

Nevertheless, existing generative studies in scoliosis have mainly focused on data augmentation or general image synthesis, rather than patient-specific postoperative prediction for surgical planning. In particular, they often lack explicit constraints to preserve spinal anatomical consistency under large corrective deformation and do not model the preoperative-to-postoperative transformation required for clinically meaningful planning support. These limitations motivate the development of a generative framework that can produce anatomically faithful postoperative images while also capturing implant-related structures relevant to surgical decision-making.

## 3. Methodology

### 3.1 Data collection and preprocessing

A total of 612 AIS patients who underwent posterior corrective fusion surgery were retrospectively enrolled from Queen Mary Hospital and Duchess of Kent Children Hospital. This study was approved by institutional review board (IRB No.: UW 25-300), with informed consent obtained from all patients and their guardians. Eligible patients were aged 10–18 years, had a clinical and radiographic diagnosis of AIS, and had complete surgery records, preoperative and postoperative spinal X-rays , and surgical planning data. Patients with other spinal deformities, previous spinal surgery or poor imaging quality were excluded.

Raw DICOM images were converted to 8-bit grayscale, standardized to 240×480 pixels, and preprocessed using adaptive contrast enhancement and Gaussian denoising (standard deviation: $\sigma$=1.0, kernel size: 3×3). Preoperative region-of-interest (ROI) masks were

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double-blinded annotated by two senior spine surgeons, and processed X-rays combined with ROI masks served as dual-channel inputs. The dataset was divided into training, validation and test sets at 8:1:1. For evaluation, ground truth (GT) annotations, including implant segmentation masks, fusion-level labels, and geometric parameters, were provided by senior spine surgeons and imaging analysts.

### 3.2 Model

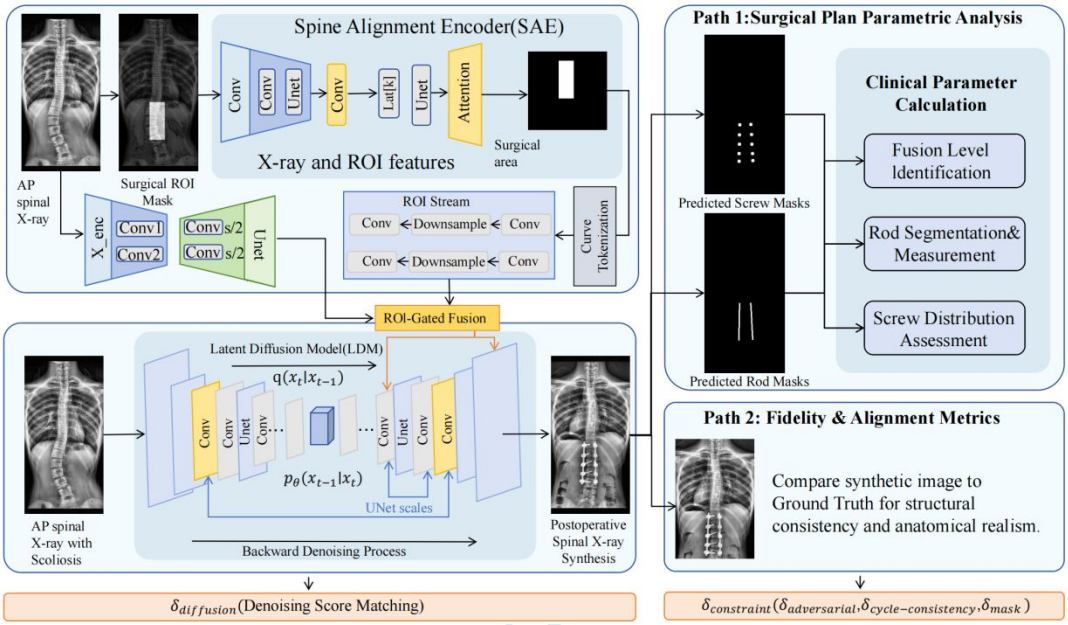



Figure 1. Overview of the proposed Surgi-Diff for surgical parameter prediction and postoperative spinal X-ray synthesis.

The overall framework of the proposed model is depicted in Figure 1.The system integrates an encoder for feature extraction from preoperative spinal X-rays, a latent diffusion model for generating synthetic postoperative images, and a two-path evaluation framework. Path 1 performs surgical parameter analysis, while Path 2 evaluates the fidelity and anatomical alignment of the synthetic images. The model uses denoising score matching and adversarial constraints for optimization.

Given the pre-operative frontal X-ray image $X_{\text{pre}} \in \mathbb{R}^{B\times1\times H\times W}$ and the ROI mask $R \in \mathbb{R}^{B\times1\times H\times W}$, this paper first concatenates the two along the channel dimension to construct a dual-channel prompt input

$$H_{\text{hint}} = \text{Concat}(X_{\text{pre}},R) \in \mathbb{R}^{B\times2\times H\times W}.$$

Subsequently, $H_{\text{hint}}$ is fed into a SAE. Within this module, the input enters an X-ray branch and an ROI branch respectively for feature extraction, obtaining multi-scale features $\{x_i\}$ and $\{r_i\}$. At each scale, ROI features are used to perform gated fusion on the X-ray features:

$$g_i = \sigma\big(\text{Conv}_{1\times1}(r_i)\big), \quad p_i = \text{Conv}_{1\times1}(r_i),$$
$$\tilde{x}_i = x_i \odot (1 + \alpha_{\text{roi}} g_i) + p_i,$$

where $\sigma(\cdot)$ denotes sigmoid function, $\odot$ denotes element-wise multiplication, and $\alpha_{\text{roi}}$ is the ROI modulation coefficient. After encoding, the SAE outputs latent space features $F_{\text{latent}}$, a spine response map $S_{\text{logit}}$, and multi-scale control features $\mathcal{C}$:

$$(F_{\text{latent}},S_{\text{logit}},\mathcal{C}) = \text{SAE}(H_{\text{hint}}).$$

Next, the spine response map $S_{\text{logit}}$ is sent to a curve tokenization module. First, a softmax operation is performed along the width direction for each row to calculate the spinal center positions:

$$P(y,x) = \frac{\exp(S_{\text{logit}}(y,x)/\tau)}{\sum_{x'}\exp\left(S_{\text{logit}}(y,x')/\tau\right)},$$

$$x_{\text{center}}(y) = \sum_{x} P\,(y,x)x.$$

Then, $N$ sampling points $\{(x_n,y_n)\}_{n=1}^{N}$ are selected along the center curve, and the corresponding features are extracted from the latent space feature map to obtain the initial tokens:

$$T_{raw} = \text{GridSample}\left(F_{latent},\{(x_n,y_n)\}_{n=1}^{N}\right) \in \mathbb{R}^{B\times N\times 256}.$$

Subsequently, through linear projection and Transformer encoding, these are mapped into structural alignment tokens:

$$T_{align} = \text{Transformer}\left(\text{Linear}\left(\text{LN}(T_{raw})\right)\right) \in \mathbb{R}^{B\times N\times 768}.$$

In the diffusion generation stage, this paper does not introduce text conditions but instead directly uses the structural alignment tokens $T_{align}$ as the context condition for the diffusion model. Meanwhile, the multi-scale control features $C$ output by the SAE are injected into a controlled U-Net to complete the latent space denoising:

$$T_{ctz} = T_{align},$$

$$\hat{\epsilon} = \text{ControlledUNet}(z_t,t,T_{ctz},C),$$

$$\hat{X}_{post} = \text{VAE}_{dec}(z_0).$$

Thus, the model generates a synthetic post-operative X-ray image $\hat{X}_{post}$. To further predict the metal implant regions, the generated image is first converted to a grayscale image $\hat{X}_{gray}$ then concatenated with the ROI mask and fed into a lightweight segmentation head:

$$X_{seg} = \text{Concat}\left(\hat{X}_{gray},R\right),$$

$$\left(\hat{M}_{screw},\hat{M}_{rod}\right) = \text{SegHead}\left(X_{seg}\right).$$

### 3.3 Evaluation metrics

In order to comprehensively evaluate the performance of the Surgi-Diff framework, this study constructed an evaluation system from two aspects: Surgical Plan Parametric Analysis and Image Fidelity and Alignment.

#### 3.3.1 Surgical plan parametric analysis

The evaluation focuses on the accuracy of the predicted internal fixation strategy and the derivation of clinically relevant parameters.

**a. Fusion level identification**

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The upper and lower boundaries of the fusion segment correspond to the Upper Instrumented Vertebra (UIV) and Lower Instrumented Vertebra (LIV), respectively. To achieve pixel-level quantitative evaluation, the center coordinates of UIV and LIV in the image are adopted as evaluation indicators instead of vertebral sequence indices. We assess the localization performance based on pixel center distance and tolerance-based accuracy:

- *Pixel Center Error* (E): For each boundary (UIV and LIV), the Euclidean distance between the predicted center coordinate $(x_{pred},y_{pred})$ and the GT center coordinate $(x_{gt},y_{gt})$ is calculated as:

$$E = \sqrt{(x_{pred}-x_{gt})^2 + (y_{pred}-y_{gt})^2}.$$

- *Accuracy* (Hit Rate): A prediction is regarded as correct when the pixel center error is less than the predefined tolerance threshold $T$. The accuracy is defined as:

$$Acc = \mathbb{I}(E \leq T),$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

**b. Rod geometric measurement**

To ensure the geometric consistency of the predicted rods, we analyze the rod centerline $C$ and endpoints $p_1,p_2$:

- *Length Error* ($E_L$): The absolute difference between the predicted and ground truth centerline lengths.
- *Endpoint Error* ($E_{end}$): The mean Euclidean distance between the predicted and ground truth endpoints.
- *Centerline Alignment Error* ($E_{cline}$): Quantified using the bidirectional Average Nearest Neighbor distance to measure spatial fitting accuracy.

**c. Screw distribution assessment**

Defining the sets of screw instance center points as $P_{pred}$ and $P_{gt}$, we evaluate:

- *Quantity Error* ($E_{\#}$): The discrepancy in the total number of predicted screws.
- *Position Error* ($E_{pos}$): The mean Euclidean distance of matched screw pairs determined via the nearest neighbor algorithm.

**3.3.2 Image Fidelity and Alignment Metrics**

To quantify the visual realism and anatomical consistency of the synthetic postoperative X-rays relative to the ground truth, the following image-level metrics are utilized:

**a. Peak Signal-to-Noise Ratio (PSNR)**

PSNR is employed to evaluate pixel-wise reconstruction quality based on the Mean Squared Error (MSE).

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX^2}{MSE+\epsilon}\right)$$

**b. Structural Similarity Index (SSIM)**

SSIM provides a holistic assessment of luminance, contrast, and structural information between the synthetic image ($I_{syn}$) and the ground truth ($I_{gt}$):

$$SSIM(I_{syn},I_{gt}) = \frac{(2\mu_s\mu_{gt}+C_1)(2\sigma_{sg}+C_2)}{(\mu_s^2+\mu_{gt}^2+C_1)(\sigma_s^2+\sigma_{gt}^2+C_2)}$$

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where $\mu$ represents the mean, $\sigma^2$ the variance, and $\sigma_{sg}$ the covariance of the respective images.

## 4. Results

### 4.1 Objective results of automatic metrics

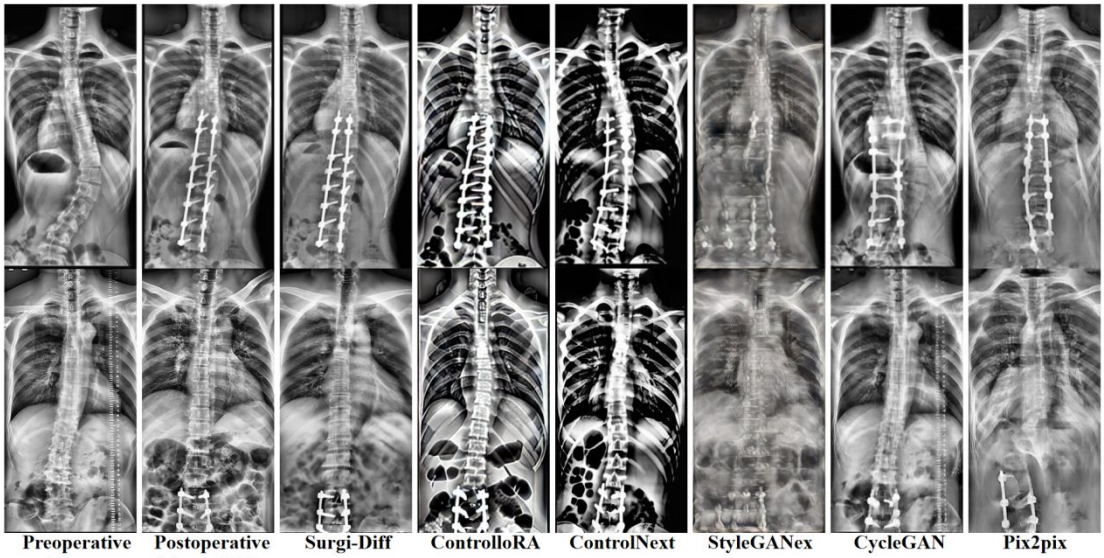



Figure 2. Qualitative comparison of synthesized postoperative X-ray results across different methods in representative AIS cases. For each case, the preoperative X-ray, ground-truth (GT) postoperative X-ray, and images synthesized by different methods are shown from left to right. Surgi-Diff generates results that more closely resemble the GT in terms of spinal alignment, postoperative morphology, and implant depiction.

Figure 2 presents qualitative comparisons of synthesized postoperative spinal X-ray across multiple methods in representative AIS cases. While all methods attempt to generate postoperative appearances from preoperative inputs, significant differences exist in structural fidelity and image realism. GAN-based approaches, such as CycleGAN and Pix2Pix, result in blurred implant structures and disconnected rods, failing to capture the precise geometry of the instrumentation. StyleGANex suffers from severe texture degradation and loss of anatomical details, making it unsuitable for clinical interpretation. Although control-based methods like ControlLoRA and ControlNext provide better spinal alignment, they introduce unnatural contrast artifacts and structural distortions, leading to a synthetic, non-clinical appearance with "shattered" implant depictions. In contrast, Surgi-Diff generates images with superior fidelity, maintaining both natural bone textures and sharp, well-defined implant structures that are highly consistent with the GT. These results demonstrate that Surgi-Diff achieves the best anatomical plausibility and geometric accuracy among all compared methods.

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Table 1. Quantitative comparison of synthesized postoperative X-ray results across different methods in terms of PSNR and SSIM.

| method | PSNR (dB) ↑ | | SSIM ↑ | |
|---|---|---|---|---|
| | Mean | SD | Mean | SD |
| Pix2pix [20] | 11.55 | 1.07 | 0.19 | 0.04 |
| CycleGAN [21] | 11.49 | 1.12 | 0.20 | 0.05 |
| ControlLoRA [22] | 10.05 | 0.88 | 0.14 | 0.03 |
| ControlNet [23] | 8.67 | 0.65 | 0.11 | 0.03 |
| StyleGANex [24] | 15.64 | 1.31 | 0.34 | 0.05 |
| Surgi-Diff | 14.25 | 2.07 | 0.35 | 0.14 |

Table 1 presents a quantitative comparison of different methods for postoperative X-ray synthesis using PSNR and SSIM. Overall, Surgi-Diff achieves competitive performance, obtaining the highest SSIM and comparable PSNR among all methods. In contrast, traditional GAN-based approaches such as Pix2pix [20] and CycleGAN [21] yield relatively lower structural similarity, indicating limited capability in preserving anatomical details. Control-based methods (ControlLoRA [22] and ControlNet [23]) further underperform in both metrics, suggesting difficulties in modeling complex postoperative structures. Although StyleGANex [24] achieves the highest PSNR, its SSIM is slightly lower than that of Surgi-Diff, implying less consistent structural reconstruction. These results demonstrate that Surgi-Diff provides a better balance between pixel-level fidelity and structural consistency.

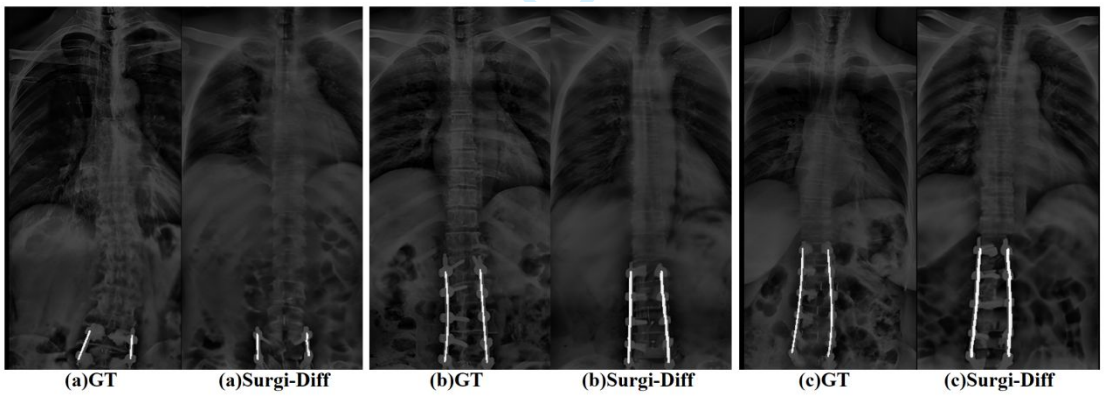



Figure 3. Qualitative evaluation of screw-rod construct prediction, comparing GT postoperative X-rays and Surgi-Diff-generated results in three representative AIS cases.

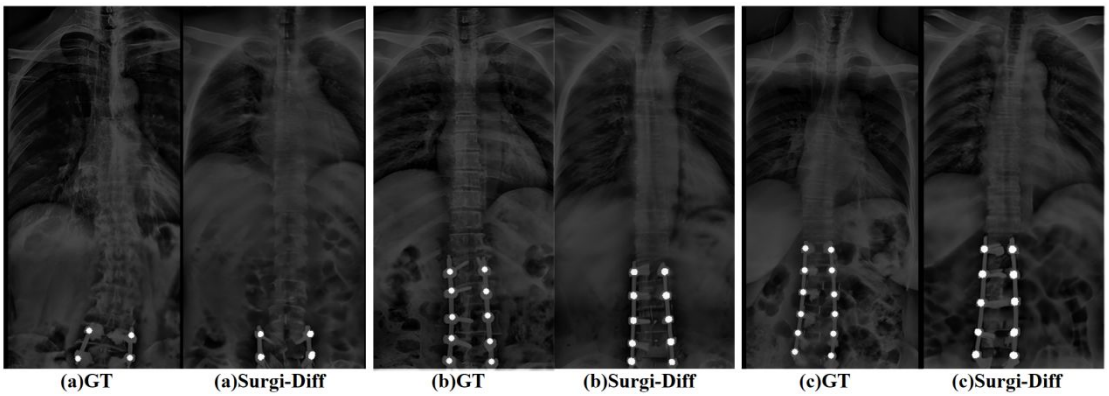



Figure 4. Qualitative evaluation of pedicle screw placement accuracy, comparing GT

postoperative X-rays and Surgi-Diff-generated results in three representative AIS cases.

Figure 3 and Figure 4 present qualitative evaluations of postoperative implant prediction in three representative AIS cases, focusing on the global screw-rod construct and the fine-grained pedicle screw placement, respectively. For each case, the GT postoperative X-ray is compared with the corresponding image synthesized by Surgi-Diff. As shown in Figure 2, the generated results demonstrate strong agreement with the ground truth in the overall morphology and bilateral configuration of the screw-rod construct, indicating that the model can reliably recover the global implant pattern. As shown in Figure 3, the predicted pedicle screw positions are highly consistent with the ground truth in terms of spatial distribution and segmental alignment along the spine, further highlighting the model's ability to capture fine-grained implant placement details. Taken together, these qualitative results suggest that Surgi-Diff can generate anatomically coherent and clinically plausible postoperative implant predictions with high structural fidelity.

Table 2. Quantitative Evaluation of Implant Prediction Performance Across Multiple Clinical Tasks.

| Category | Metric | Symbol | Result |
|---|---|---|---|
| Fusion Level Identification | Accuracy | Acc | 75.8% |
| | Boundary Index Error(UIV) | $E_{UIV}$ | 8.40 pixel |
| | Boundary Index Error(LIV) | $E_{LIV}$ | 7.88 pixel |
| Rod Geometry Measurement | Length Error | $E_L$ | 19.36 pixel |
| | Endpoint Error | $E_{end}$ | 4.50 pixel |
| | Centerline Error | $E_{cline}$ | 2.41 pixel |
| Screw Distribution | Quantity Error | $E_{\#}$ | 1.09 |
| | Position Error | $E_{pos}$ | 5.49 pixel |

Table 2 presents the quantitative results of Surgi-Diff across several clinical evaluation metrics. In fusion level identification, the model achieves an accuracy of 75.8% at a 10-pixel tolerance, with the boundary index errors for UIV ($E_{UIV}$=8.40) and LIV ($E_{LIV}$=7.88) both remaining under 10 pixels. For rod geometry measurement, the model demonstrates high precision in capturing the rod's trajectory, as evidenced by a low centerline error of 2.41 pixels, although a relatively higher length error ($E_L$=19.36) is observed. Regarding screw distribution, Surgi-Diff yields a mean quantity error of 1.09 and a position error of 5.49 pixels, indicating a reasonable capability in localizing spinal implants. Overall, these findings suggest that Surgi-Diff provides a robust foundation for automated spinal instrumentation analysis.

**5. Limitations**

This study has several limitations. First, the PSNR and SSIM values in Table 1 are not yet optimal. A possible reason is that postoperative radiograph synthesis is not a strictly deterministic task, and multiple plausible postoperative appearances may correspond to the same preoperative input. Therefore, conventional pixel-wise metrics may not fully reflect the

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anatomical plausibility or clinical usefulness of the generated results. Although further improvement in image fidelity is still needed, the qualitative findings and implant-related analyses suggest that the proposed framework already provides meaningful structural predictions that may support preoperative planning.

## 6. Conclusion

We proposed Surgi-Diff, a diffusion-based framework for postoperative outcome prediction and surgical planning in AIS. By integrating structural priors from preoperative X-rays and surgical region masks, Surgi-Diff generates anatomically consistent postoperative images and predicts implant-related structures with favorable realism and clinical utility. These results suggest its potential value for AIS preoperative planning, outcome visualization, and surgical decision support.

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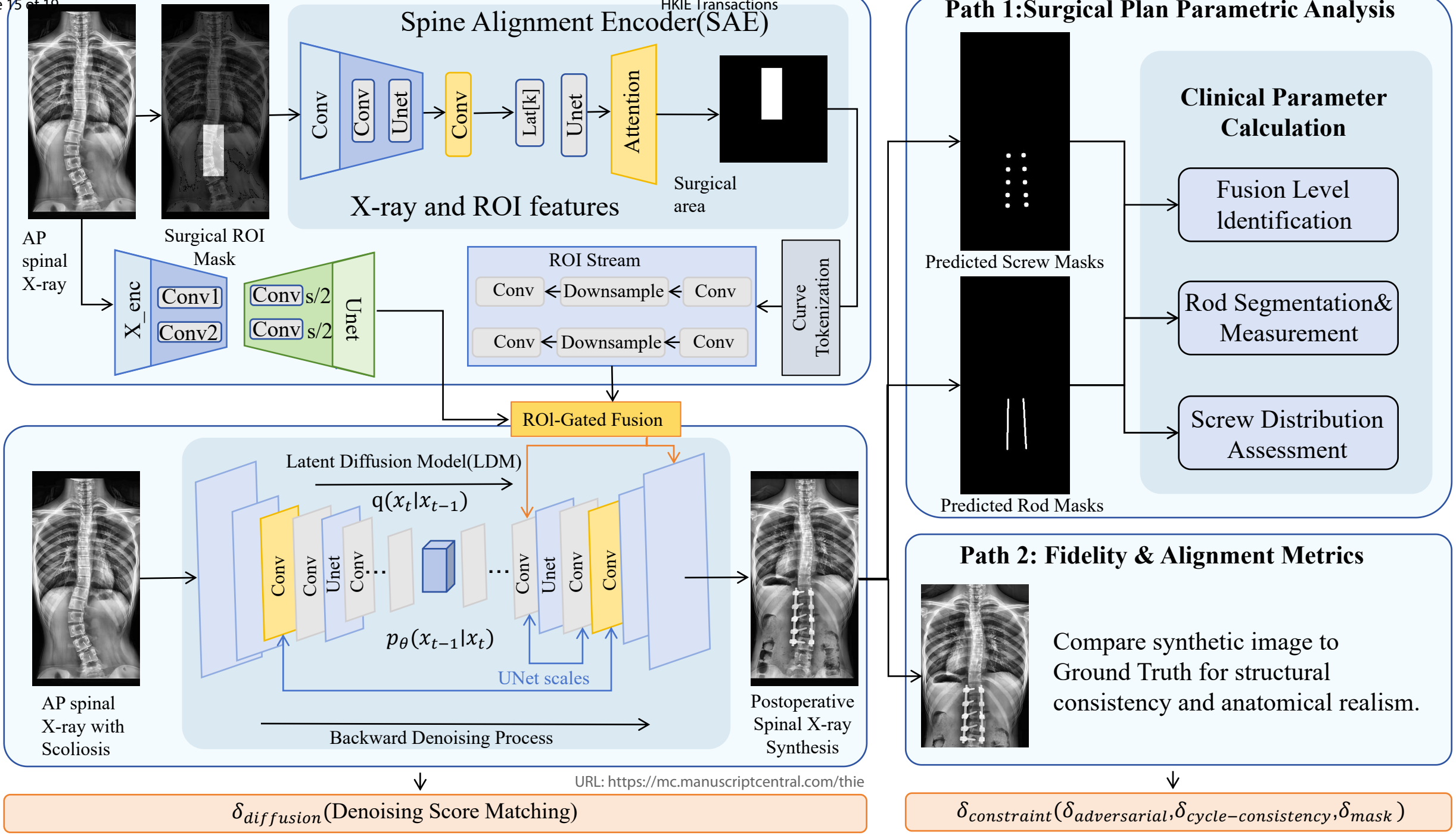
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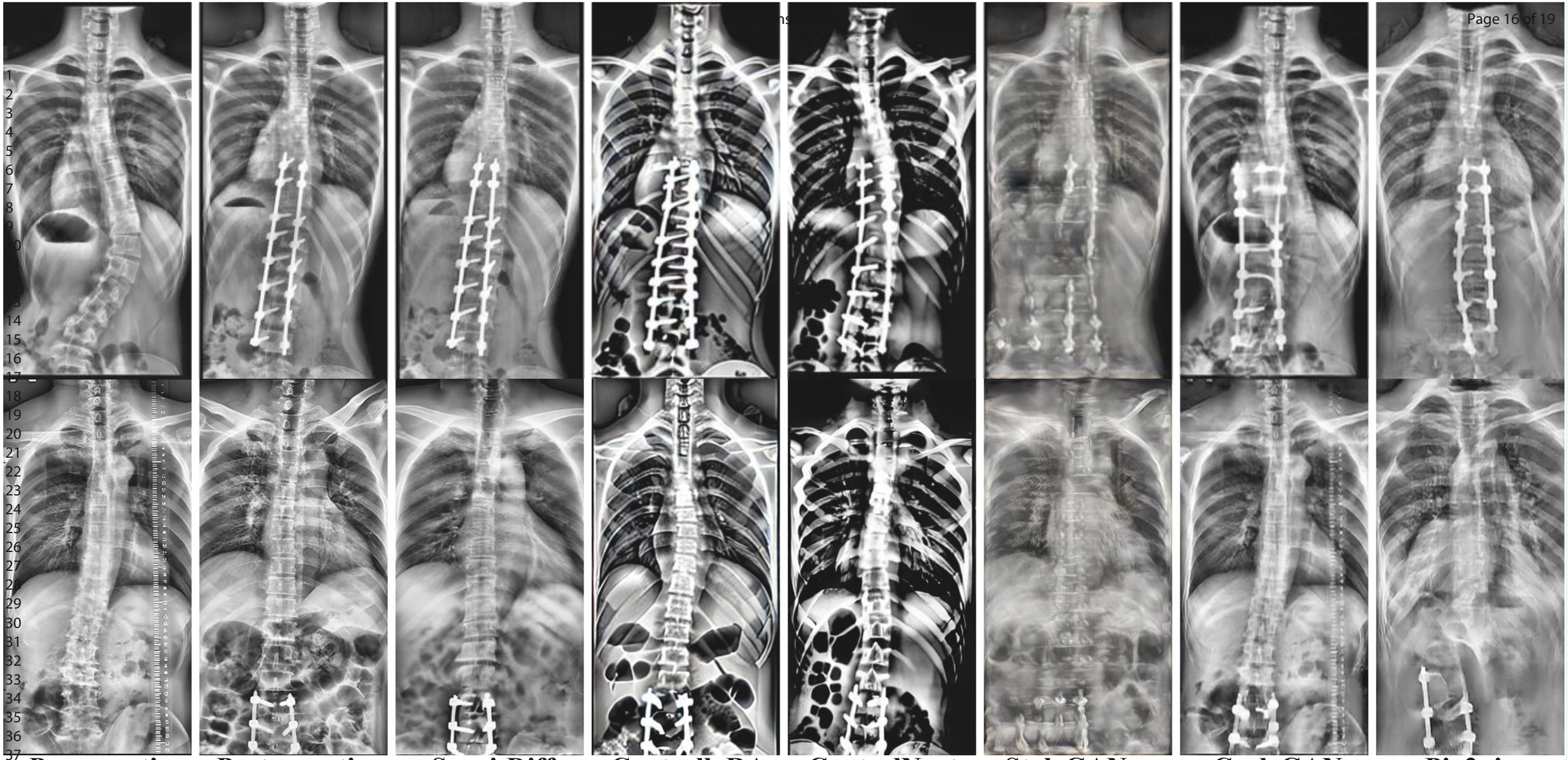
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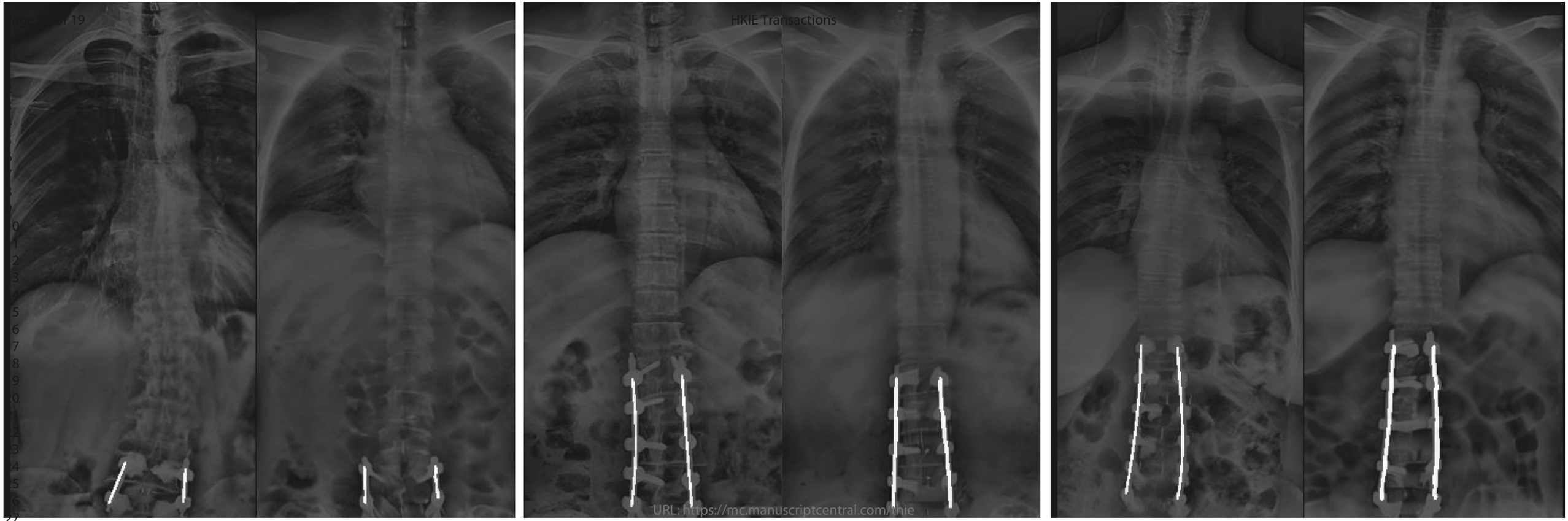
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Spine Alignment Encoder(SAE)
AP spinal X-ray
Surgical ROI Mask
Conv
Conv
Unet
Conv
Lat[k]
Unet
Attention
Surgical area
X-ray and ROI features
X_enc
Conv1
Conv2
Conv s/2
Conv s/2
Unet
ROI Stream
Conv
Downsample
Conv
Conv
Downsample
Conv
Curve Tokenization
ROI-Gated Fusion
Latent Diffusion Model(LDM)
$q(x_t|x_{t-1})$
$p_\theta(x_{t-1}|x_t)$
UNet scales
Backward Denoising Process
AP spinal X-ray with Scoliosis
Postoperative Spinal X-ray Synthesis
Path 1:Surgical Plan Parametric Analysis
Predicted Screw Masks
Predicted Rod Masks
Clinical Parameter Calculation
Fusion Level Identification
Rod Segmentation& Measurement
Screw Distribution Assessment
Path 2: Fidelity & Alignment Metrics
Compare synthetic image to Ground Truth for structural consistency and anatomical realism.
$\delta_{diffusion}$(Denoising Score Matching)
$\delta_{constraint}(\delta_{adversarial}, \delta_{cycle-consistency}, \delta_{mask})$

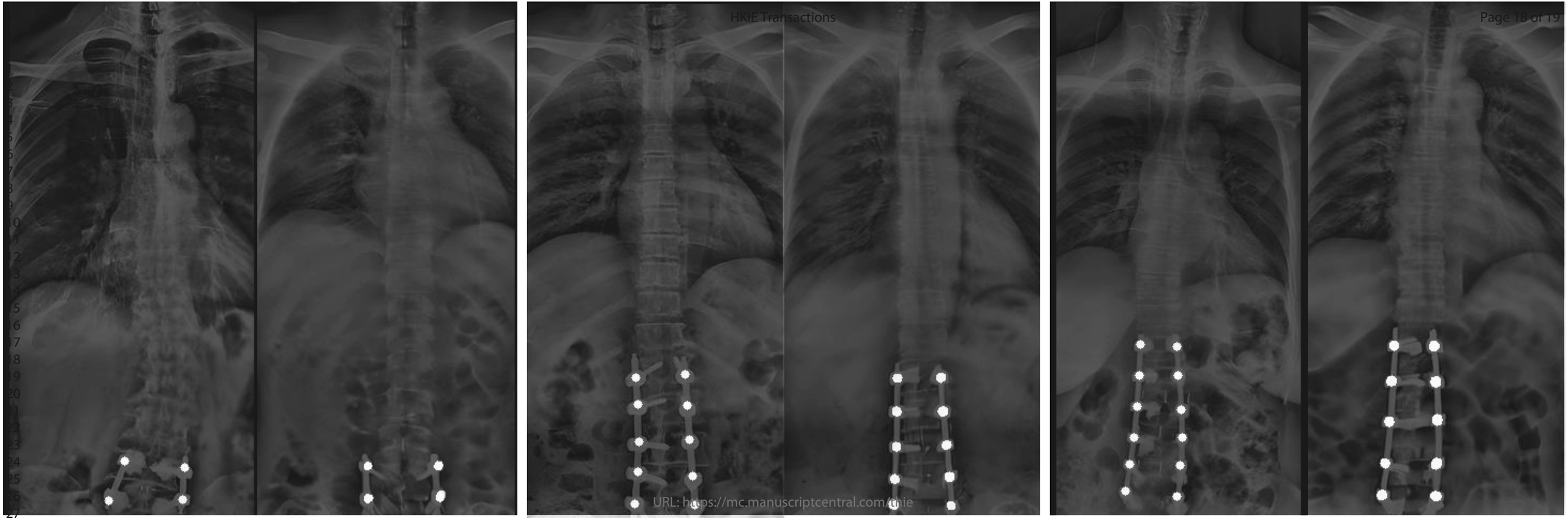



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Table 1: Quantitative comparison of postoperative X-ray synthesis performance across different methods in terms of PSNR and SSIM.

| Method | PSNR (dB)↑ | | SSIM↑ | |
|---|---|---|---|---|
| | Mean | SD | Mean | SD |
| Pix2Pix | 11.55 | 1.07 | 0.19 | 0.04 |
| CycleGAN | 11.49 | 1.12 | 0.20 | 0.05 |
| ControlLoRA | 10.05 | 0.88 | 0.14 | 0.03 |
| ControlNet | 8.67 | 0.65 | 0.11 | 0.03 |
| StyleGANex | 15.64 | 1.31 | 0.34 | 0.05 |
| Sargi-Diff | 14.25 | 2.70 | 0.35 | 0.14 |

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Table 1: Quantitative Evaluation of Implant Prediction Performance Across Multiple Clinical Tasks

| Category | Metric | Symbol | Result |
|---|---|---|---|
| Fusion Level Identification | Accuracy | Acc | 75.8% |
| | Boundary Index Error (UIV) | $E_{\text{UIV}}$ | 8.40 pixel |
| | Boundary Index Error (LIV) | $E_{\text{LIV}}$ | 7.88 pixel |
| Rod Geometry Measurement | Length Error | $E_L$ | 19.36 pixel |
| | Endpoint Error | $E_{\text{end}}$ | 4.50 pixel |
| | Centerline Error | $E_{\text{cline}}$ | 2.41 pixel |
| Screw Distribution | Quantity Error | $E_{\#}$ | 1.09 |
| | Position Error | $E_{\text{pos}}$ | 5.49 pixel |

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